SURPLUS SERVE – Bridging Feast to Need

Submitted in partial fulfilment of the requirements of the degree of

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CERTIFICATE

This is to certify that the project entitled "**Surplus Serve – Bridging Feasts to Need**" is a Bonafide work of **Yash Sawant (23106009), Sohan Vernekar (23106012), Parth Rane (23106018)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (AI&ML)**

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Project Report Approval for T.E.

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Abstract

Food wastage is a global issue of utmost importance, with nearly one-third of the total food produced being lost or wasted when millions are undernourished and hungry. India alone has lavish events like weddings, corporate functions, and festivals that produce huge amounts of excess food, which frequently lay untouched because of the inefficient coordination mechanisms in place between event planners and charity organizations. This project offers the design, development, and deployment of an NGO Food Donation App: a holistic mobile-based platform that connects the food donor with non-governmental organizations working among disadvantaged populations.

The app is developed with Flutter framework for cross-platform mobile application development and is compatible with both Android and iOS devices and utilizes Firebase cloud services for backend support such as authentication, real-time syncing of the database, cloud messaging, and secure storage. The platform adopts role-based structure differentiating between two major types of users: event organizers who create announcements of remaining food, and NGO officials who are notified in real-time to ease prompt collection and distribution. Major technical features are Firebase Cloud Messaging for real-time push notifications based on geographical location, integration of Google Maps API for location-based NGO finding, Cloud Firestore for real-time data synchronization negating manual refresh processes, an inbuilt bidirectional messaging component allowing real-time coordination between organizers and NGOs.

Through simulated case studies and projected metrics, the system demonstrates potential for reducing food wastage by 70–90% per participating event while serving an average of 250–600 beneficiaries per donation. The platform addresses operational inefficiencies by reducing coordination time from hours to under 30 minutes, improves NGO resource utilization through optimized logistics, and provides measurable environmental benefits including carbon footprint reduction and water resource conservation.

This study adds to a scalable, technology-based solution that turns the consumption of food donations from an informal and manual system into a structured, efficient, and transparent system. The result illustrates that the careful use of modern technology can generate high social benefit by rectifying coordination failures that contribute to undue wastage of food in the face of existing food insecurity.

Sustainability Goals Mapped: SDG2, SDG3, SDG11, SDG12, SDG13, SDG17

Keywords: Food Waste Reduction, Mobile Application Development, Flutter Framework, Firebase Cloud Services, Real-Time Notification Systems, Location-Based Services, Social Impact Technology, NGO Management, Donation Coordination, Cross-Platform Development

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ABBREVATION

AI - *Artificial Intelligence*

ML - *Machine Learning*

API - *Application Programming Interface*

FCM - *Firebase Cloud Messaging*

GPS - *Global Positioning System*

HTTP - *Hypertext Transfer Protocol*

JSON - *JavaScript Object Notation*

NGO - *Non-Governmental Organization*

NoSQL - *Not Only Structured Query Language*

REST - *Representational State Transfer*

SDK - *Software Development Kit*

UI - *User Interface*

Chapter 1

Introduction

1.1 Introduction

Food wastage is among the world's most pertinent social, economic, and environmental issues. According to the Food and Agriculture Organization (FAO), almost 1.3 billion tons of food, or one-third of all that is produced worldwide, is lost or wasted annually. This becomes even more precarious when juxtaposed against the reality that more than 820 million individuals live chronically hungry. In India alone, more than 194 million individuals are still undernourished, and the food-hunger paradox is worst here.

Mass social gatherings like festivals, weddings, and corporate events are major contributors to food wastage. Weddings in India generally feed 20–30% more people than invited, resulting in huge leftovers. The same over-preparation is done for other events in order to prevent shortages. Although the genesis of the action has cultural overtones of hospitality, the outcome

is huge excess food that usually ends up being wasted because there are no efficient redistribution mechanisms in place.

The environmental impact is also worrying. Food waste that has decomposed in landfills releases methane, a 25 times more powerful greenhouse gas than carbon dioxide. Food waste also means lost resources in the form of water, energy, and fertilizers employed in production. Research approximates that food waste accounts for about 8–10% of all global greenhouse gas emissions.

From a social perspective, the presence of food wastage alongside hunger highlights a failure of coordination at the systemic level. While numerous non-governmental organizations (NGOs) and community kitchens endeavor to fight hunger, they may struggle to access adequate food supplies. The primary problem is not a lack of intent to donate or accept food but an inefficiency in systems to link surplus donors (event organizers) and recipients (NGOs).

Conventional donation procedures are slow and inefficient. Organizers have to scout around local NGOs, call them up, arrange details, and arrange pickups—through hectic post-event times. NGOs, conversely, depend on word of mouth or personal referrals for hearing about donations available to them, usually too late in the hour. This results in missed opportunities and wastage.

With advances in smartphone technology and cloud applications, digital platforms have revolutionized industries by linking users in an efficient manner ride-sharing and food delivery apps are good examples. Translating the same technological concepts to food donation can bridge the gap between donors and receivers, minimize waste, and maximize social return.

1.2 Problem Identification

The existing food donation landscape has some key problems:

Information Asymmetry: Event planners do not have access to a single platform to identify nearby NGOs that can take food donations. Likewise, NGOs also do not have real-time insight into where and when there is excess food available.

Coordination Complexity: Even after establishing contact, manual coordination on food type, volume, pickup time, and logistics causes delays. Considering that cooked food can only remain safe for a few hours, these delays oftentimes render the donation impossible.

Logistical Limitations: NGOs often do not have enough transport or volunteers to pick up food on short notice, whereas organizers cannot cope with deliveries after an event. This coordination gap leads to missed opportunities.

Trust and Verification Issues: Organizers fear exploitation of food or legal repercussions, whereas NGOs are concerned about the quality of food and the reliability of donors. In the absence of verification systems, trust is a dominant obstacle.

Absence of Documentation and Impact Tracking: Informal food donation systems seldom keep track of food quantities received or numbers of people served. Lack of data constrains accountability, measuring impacts, and planning ahead.

1.3 Project Objectives

The key aim of this project is to devise and implement a modular, cloud-based mobile application which helps support effective, transparent, and scalable coordination among NGOs, volunteers, and donors who are engaged in food donation efforts. The system will help reduce food wastage, improve communication between the stakeholders, and ensure that excess food reaches the needy population in a timely and organized fashion.

The following are the specific objectives, which direct the system development and implementation:

1. **Role-Based Modular Mobile Platform Development**: Design a mobile app supporting three main user roles NGO Administrator, Volunteer, and Donor with each

having a specific dashboard and tailored functionality. The NGO Administrator module will manage event scheduling, campaign monitoring, and performance reporting. The Volunteer module will have event registration, participation monitoring, and live updates. The Donor module will enable users to trigger and track donations, see contribution effect, and get thanks.

2. **Implement Secure Authentication and Role-Based Access Control (RBAC):** Design and incorporate a secure authentication mechanism utilizing Firebase Authentication to handle user identity and access to system modules. Each user will be assigned permissions based on their role, which will maintain the confidentiality and integrity of data as well as restrict unauthorized access.
3. **Facilitate Real-Time Data Synchronization and Communication**: Utilize Firebase Firestore to provide real-time synchronization of donation information, event status, and volunteer activities between all devices. In-app messaging or notification systems integrated within will immediately inform users of new campaigns, food stocks available, or assignment of tasks, reducing response time and enhancing coordination.
4. **Incorporate Cloud-Based Infrastructure for Scalability**: Use Firebase Cloud Services as the backend infrastructure for scalability, reliability, and low server maintenance. The cloud-based architecture will have the ability to accommodate a growing user base so that the application is able to scale across different NGOs and geospatial locations without any significant architectural modifications.
5. **Offer Location-Aware Features and Event Tracking**: Embed Google Maps API to support geolocation-based functions like locating nearby donation points, delivery route monitoring, and volunteer assignment optimization. This helps in smooth logistical coordination, particularly during massive events or emergency food distribution campaigns.
6. **Encourage Social Responsibility with Technology**: Apart from technical objectives, the project aims to develop a platform that fosters community engagement, food wastage awareness, and collaborative work toward the attainment of sustainable development goals (SDG 2 – Zero Hunger, SDG 12 – Responsible Consumption and Production).

The project involves the designing, developing, and deployment of a cross-platform mobile application that connects to the cloud, simplifies food donation operations, and enables coordination among NGOs, volunteers, and donors. The system integrates modular role-based dashboards, enabling each stakeholder to access dedicated functionalities tailored to their responsibilities NGOs can create and manage donation campaigns, volunteers can register for events and track participation, and donors can contribute food or funds while monitoring the impact of their contributions. Developed with Flutter as the frontend and Firebase as the backend technology stack, the app provides real-time syncing, secure authentication, and smooth user experience in all modules.

The operational scope of the project involves adopting major functionalities like event management, live tracking, notification systems, and role-based access control to be backed by cloud storage and data analytics for improved transparency and accountability. Integration with Google Maps API facilitates location-based services for the optimization of food collection and distribution routes. The application initially focuses on urban and semi-urban areas where wastage of food is high and NGOs are actively working, with the possibility of future extension to other areas. As the system addresses digital coordination and resource optimization, future releases can add more sophisticated functionalities like AI-based demand forecasting, international language support, and payment processing to further amplify its influence and reach.

Chapter 2

Literature Survey

2.1 History

The effort to rationalize food redistribution and enhance donor-NGO coordination has come a long way in recent decades. Much of the development has been influenced by rising global interest in food wastage, hunger, and lack of effective communication between donors and social agencies. Initial systems were mainly based on paper-based reporting and direct networking, which often resulted in delays, duplication of work, and inefficient utilization of available food supplies. The path towards an integrated digital food donation system has developed in conjunction with advances in information technology, cloud computing, and real-time data communication.

Early Steps (Pre-2010s):

In the initial stages, food donation and NGO coordination were largely manual. Restaurant owners and event organizers would usually have to make phone calls, rely on local word of mouth, or physically visit NGOs. Information regarding excess food, availability, and transportation logistics was stored in spreadsheets or by hand.

Whereas these approaches were delivered basic functionality, they had problems with scalability, transparency, as well as delays in communication, which would end up in wastage of perishable food items before it reached the vulnerable populations. Other early digital experiments included static web portals through which NGOs registered but which were non-interactive, not updated in real time, nor were location-sensitive.

Shift to Web-Based Platforms (2010s):

The 2010s saw the introduction of cloud-based and web-integrated platforms to make NGO functions easier and to track donations. MySQL, PHP, and some early versions of Firebase enabled NGOs to create centralized donor–NGO coordination databases.

Still, these systems were subject to manual verification and scheduling, with little automation or knowledge of their location. Most of the solutions catered to individual organizations rather than for widespread collaboration among multiple event organizers or NGOs. In spite of these issues, this period was a major milestone toward digital transparency and data being stored centrally, laying the groundwork for intelligent systems.

Modern Era (2020s–Present):

The speedy evolution of geolocation technologies, cloud computing, and mobile applications has revolutionized the way NGOs and donors connect. Present systems use Firebase for real-time synchronization, Google Maps API for location-based matching, and push notification systems for real-time communication. These technologies allow donors to post available food in real-time after events and enable NGOs to reply in real time according to proximity and requirement.

In addition, platforms such as Flutter and React Native enabled the development of cross-platform mobile applications with easy-to-use interfaces to make access easy for even those who lack advanced technical knowledge. Authentication and security processes driven by

Firebase ensure secure data management, while cloud architecture allowing scalability facilitates expansion to several cities and organizations.

The NGO Connect Application is the pinnacle of these technological developments combining real-time data exchange, geolocation monitoring, and cloud scalability into one singular, cohesive environment. Through bringing together these innovative tools, the platform not only increases operational efficacy but also achieves social sustainability by reducing wastage of food and distributing it in an equitable manner.

2.2 Related Research

Evidence on food wastage identifies its worldwide magnitude in both developed and developing countries. An estimated 13% of the world's food is lost between the harvesting and retail levels, and another 17% is wasted at the consumer level, such as in restaurants, households, and retail.

Historical food recovery efforts spearheaded by nonprofit organizations have resulted in positive outcomes but have scalability limitations. Manual coordination of phone calls, word of mouth, and ad-hoc systems causes friction and delays in collection, with resultant spoilage due to ready-to-eat food having limited safe storage life without refrigeration.

Scholarly research that analyzes behavioral and organizational obstacles to food donation finds that organizers of events are open to donating excess food but are unaware of recipient groups or fearful of legal consequences. Charities, however, find it difficult to find opportunities for donations and mobilize assets in a timely fashion. These findings indicate that technological intermediation may facilitate overcoming information asymmetry and coordination failures to enable greater food recovery.

Studies also indicate that mobile applications have the ability to effectively link providers and beneficiaries across different fields like healthcare, education, and disaster response. The keys to success are intuitive interfaces, real-time alert systems for rapid response, and trust features such as verified user profiles and feedback mechanisms.

In the food donation sector, websites such as OLIO (UK) and Food Rescue US (North America) show the way for leveraging technology to expand food sharing networks. Assessments of these platforms underscore that notifications enhance response rates, place-based matching streamlines logistics, and embedded messaging boosts coordination reliability.
Studies on user adoption of social impact applications emphasize that users expect the same level of performance and design quality as commercial apps. Hence, ensuring robust security, data privacy, and transparent operations is critical to building trust and long-term engagement.

Lastly, the technical literature for building mobile apps documents the evolution from native platforms to cross-platform platforms, which enable code reuse and quick deployment. Google's Flutter, which was released in 2018, is a prime example, providing a contemporary cross-platform development platform based on the Dart programming language.

2.3 Reference

[1] Ms. V.B.Aswale, Urmila Rajendra Shinde, Anushka Abhijit Patil, Prithviraj Vijay Sapkal, Prathmesh Bajrang Deshmukh, Nikhil Audumbar Chendage, "Helping Hand" Wastage Reductions

"Helping Hand" is an Android-based application designed to streamline donations of food, clothing, and other essentials. It connects donors with NGOs and beneficiaries through a mobile-friendly platform that tracks donations, manages inventory, and ensures efficient distribution. The app features real-time tracking, AI- powered donation recommendations, and automated reporting to optimize the donation process

[2] Aman Singh, Bocha Sahith, Likhith M, Maddineni Pallavi, "Nourishing neighbourhoods: the Impact of food bridge on local food access"

The purpose of this project is to generate a food bridge where the Function halls and NGOs will get connected through design thinking concepts. Food bridge acts as an avenue for socializing, teaching as well as sharing meals with others thus creating fairness and strength in feeding systems that can withstand any calamity. Through creative designs in mind bending techniques such as those employed by illusionists during their performances, this paper aims

at fostering equality within communities while responding towards hunger challenges experienced globally especially among children who are vulnerable due to malnutrition

[3] Ms. Pooja Vajpayee, Shiva Shukla, Rohit Kumar, "Aahar App"

The Aahar app is a food-denoting mobile application designed to help users discover and explore a wide range of food options. The app provides a platform for users to search for restaurants, cafes, food trucks, and other dining establishments based on their location, preferences, and dietary restrictions. Users can browse through a curated collection of menus, view food images and descriptions, and read reviews from other users to make informed decisions.

[4] Prof. Devachand Chaudhari, Rutika Rahul Kamble, Kunal Suraj Lalwani, Sakshi Bhaskar Pawade, Purva Satish Bondgulwar, Rutika Dipak Tonge, "Waste Food Management and Donation App"

The idea behind over project can be use by many people who wish to donate things to needy organizations Also, many organizations like to ask for various things required by them such as clothes, food grains, books utensils. In this mobile app, we have tried to reduce food wastage by giving waste food to people or organization who need it.

Chapter 3

Limitation of Existing system

1. Geographical Limitations

- Localized Reach: The app may currently operate only in specific cities or regions, limiting its ability to serve NGOs or event organizers outside those areas.
- Lack of Real-Time Location Tracking: Without precise geolocation or route optimization, matching nearby NGOs to events might be inefficient or inaccurate.

2. User Onboarding and Verification

- Manual NGO/Event Manager Verification: If verification of NGOs or event managers is done manually, it slows down onboarding and introduces trust issues.
- Incomplete Profiles: Users may skip entering critical details (e.g., storage capacity, dietary restrictions accepted, pickup windows), reducing match quality.

3. Matching and Notification Inefficiencies

- Static Matching Logic: The app might use basic rules rather than dynamic factors like NGO capacity, food type compatibility, or urgency.
- Delayed Notifications: If push notifications aren't real-time or reliable, NGOs may miss time-sensitive surplus food opportunities.
- No Auto-Matching or Bidding System: Event managers may have to manually contact multiple NGOs, leading to delays or food spoilage.

4. Food Safety and Compliance Gaps

- Lack of Food Safety Guidelines Integration: The app may not guide users on safe handling, packaging, or transportation of surplus food, risking health violations.
- No Compliance Tracking: Absence of features to log food origin, temperature logs, or donation receipts may hinder legal compliance (e.g., with Good Samaritan laws).

5. Scalability and Performance

- Backend Bottlenecks: If the backend isn't optimized for high concurrency (e.g., during large festivals with many simultaneous surplus reports), the system may slow down or crash.
- Limited Data Analytics: Without dashboards or usage analytics, it's hard to identify trends, optimize operations, or demonstrate impact to stakeholders.

6. User Experience (UX) and Accessibility

- Clunky UI/UX: Complex workflows for reporting surplus food or accepting donations may discourage regular use.
- Language or Literacy Barriers: If the app supports only one language or assumes high digital literacy, it excludes potential users (e.g., small NGO volunteers).
- No Offline Support: Event venues may have poor connectivity; without offline functionality, users can't log surplus until later by which time food may be spoiled.

7. Trust and Accountability

- No Feedback or Rating System: Lack of mutual reviews between NGOs and event managers reduces accountability and trust.
- Absence of Donation Proof: NGOs may need official donation receipts for tax or reporting purposes missing this feature reduces incentive to participate.

8. Sustainability and Maintenance

- Dependence on Volunteers: If the system relies heavily on manual coordination (e.g., admins assigning NGOs), it doesn't scale sustainably.
- Limited Integration with External Systems: No APIs to connect with logistics providers (e.g., delivery services), calendar apps, or government food donation platforms.

9. Legal and Regulatory Constraints

- Unclear Liability Framework: The app may not clarify who is liable if donated food causes illness—discouraging participation from risk-averse event managers.
- Regional Legal Variations: Food donation laws differ by country/state; a one-size-fits-all app may not comply everywhere.

10. Impact Measurement

- No Metrics on Waste Reduced or Meals Saved: Without built-in impact tracking, it's hard to attract funding, partners, or user engagement.

Chapter 4

Problem Statement and Objective

4.1 Problem Statement

Contemporary Non-Governmental Organizations (NGOs) face significant operational challenges in managing their philanthropic activities, particularly in the domains of donation management, volunteer coordination, and event organization. The absence of integrated technological solutions has resulted in fragmented workflows, diminished transparency, and suboptimal resource utilization across the nonprofit sector. Traditional approaches to NGO management rely heavily on manual processes, disparate systems, and paper-based documentation, which inherently limit scalability and operational efficiency.

The fundamental challenge lies in developing a comprehensive technological framework that addresses the multifaceted requirements of modern NGO operations while ensuring

accessibility, transparency, and accountability. Current market solutions often provide isolated functionalities, requiring organizations to implement multiple platforms for different operational aspects, thereby increasing complexity and operational overhead. Furthermore, the lack of real-time data synchronization and analytics capabilities hinders NGOs' ability to make informed strategic decisions and demonstrate impact to stakeholders.

This project aims to address these systemic inefficiencies by designing and developing a unified mobile application platform that seamlessly integrates donation management, volunteer coordination, and event organization functionalities. The proposed solution must accommodate the diverse operational requirements of NGOs while maintaining user-centric design principles and ensuring robust security measures for financial transactions and sensitive data handling.

4.2 Objectives

Building upon the foundational requirements identified in Chapter 1.3, this project establishes comprehensive technical objectives that encompass both functional and non-functional requirements essential for developing a robust NGO management platform.

The primary objective involves creating an interactive cross-platform mobile interface utilizing the Flutter framework, leveraging its declarative UI paradigm and single codebase philosophy to ensure consistent user experiences across iOS and Android platforms. Flutter's widget-based architecture and Material Design implementation provide the necessary foundation for developing responsive, accessible interfaces that accommodate diverse user demographics and technological proficiencies prevalent in the nonprofit sector.

The architectural foundation necessitates designing cloud-based data models optimized for scalability and high availability, implementing Firebase's NoSQL document database structure to accommodate dynamic data schemas and fluctuating user loads. This approach ensures horizontal scalability while maintaining data consistency across distributed systems, crucial for supporting growing NGO operations and expanding user bases.

Security and financial integrity represent critical requirements, mandating the integration of secure payment gateways that comply with industry standards such as PCI DSS for donation handling. The implementation must incorporate multi-layered security protocols, including end-to-end encryption, tokenization mechanisms, and fraud detection algorithms to protect sensitive financial information and maintain donor confidence.

Real-time data synchronization capabilities constitute another fundamental objective, implemented through Firebase's Firestore real-time database architecture. This cloud-native solution enables instantaneous updates across all connected clients, ensuring data consistency and enabling collaborative workflows essential for effective volunteer coordination and event management.

The final technical objective encompasses implementing comprehensive analytics and visualization capabilities, empowering NGOs with data-driven insights for performance monitoring and strategic decision-making. This involves developing dashboard interfaces that aggregate key performance indicators, donation trends, volunteer engagement metrics, and event participation statistics, utilizing visualization libraries to present complex data in accessible, actionable formats that support organizational growth and impact measurement.

Chapter 5

Proposed System

5.1 System Overview

The proposed system constitutes a modular, cloud-connected mobile application engineered to streamline food donation processes and facilitate seamless collaboration among three primary stakeholder roles: the NGO administrator, the volunteer, and the donor. Each role is provisioned with a dedicated, role-specific dashboard interface that presents contextually relevant features and data visualizations. The NGO administrator's dashboard enables comprehensive event management capabilities, including campaign creation, tracking of donation progress, and generation of performance reports. Volunteers access a curated list of upcoming and ongoing events, with functionality to register for participation, receive real-time updates, and communicate with event coordinators. The donor dashboard offers an intuitive interface for initiating donations, monitoring the impact of contributions through interactive summaries, and obtaining receipts and acknowledgments. This modular approach ensures that each user role encounters a tailored experience, enhancing usability while

maintaining a shared data structure and unified cloud backend. Role-based access control guarantees data security and privacy, with authentication and authorization mechanisms governing access to each module's operations.

The frontend is implemented in Flutter, leveraging Dart to deliver a performant, cross-platform mobile application. Flutter's widget-based architecture facilitates rapid UI prototyping and ensures consistent rendering across Android and iOS devices. On the server side, Firebase Firestore serves as the primary NoSQL document database, enabling flexible schema design and real-time data synchronization. Firebase Authentication handles secure user sign-in, supporting email/password credentials and third-party OAuth providers.

5.2 Methodology

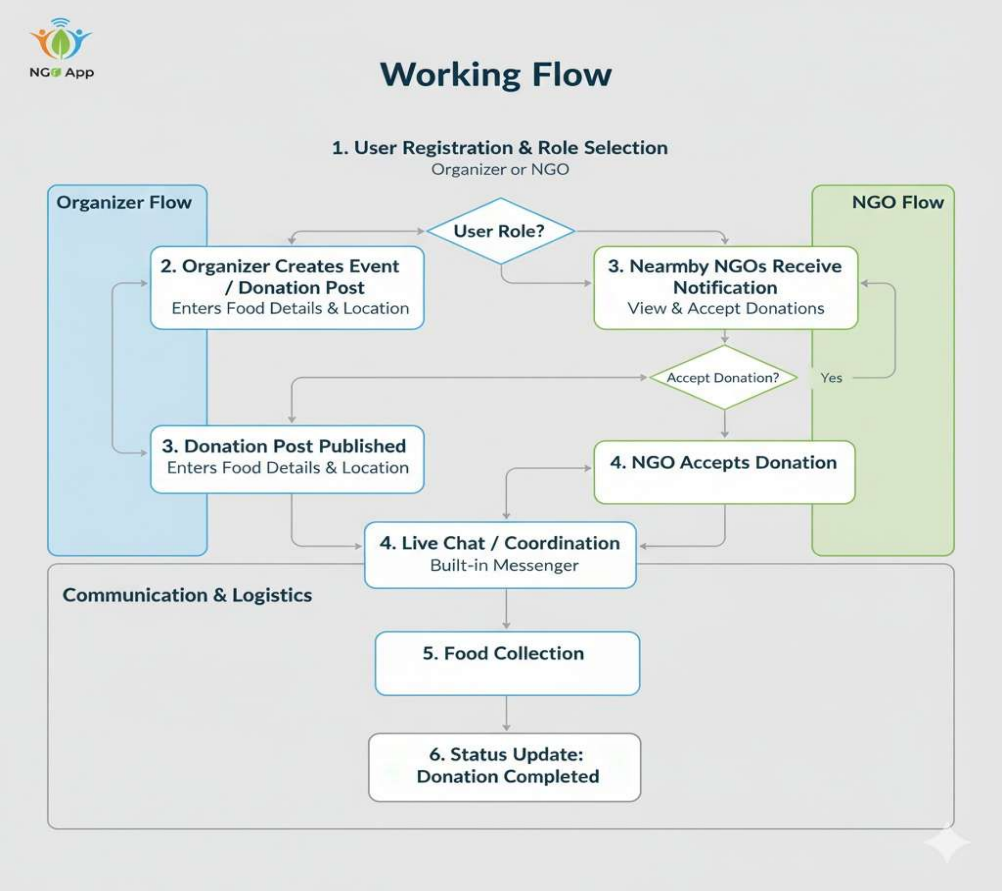

Fig 5.1 Workflow

Working Flow Explanation

The working flow diagram illustrates the operational sequence and data exchange between the Organizer (Donor) and the NGO within the proposed food donation application. It highlights the systematic process from user registration to completion of the donation, integrating both communication and logistical functions. The flow is divided into three main segments: Organizer Flow, NGO Flow, and Communication & Logistics.

1. User Registration and Role Selection

The process begins with the registration of a new user, who must choose their role as either an Organizer (representing individuals, event coordinators, or donors) or an NGO (a charitable organization responsible for receiving and distributing food donations).

This step ensures role-based access control within the system, where each user is directed to functionalities and dashboards specific to their role.

Once the role is selected, the workflow diverges into two distinct pathways — Organizer Flow and NGO Flow — each governing separate sets of activities

2. Organizer Creates Event / Donation Post

If the user selects the Organizer role, they are directed to create a new donation post or event. In this step, the organizer enters essential details such as Type and quantity of food available, Expiry or collection time, Pickup location, Any additional notes regarding packaging or delivery.

This information forms the basis of the donation listing, which will later be visible to nearby NGOs through the system's cloud backend.

3. Donation Post Published / Nearby NGOs Receive Notification

After the donation post is created, the system automatically publishes it to the cloud database (Firebase Firestore). The backend then triggers a real-time notification to all nearby registered NGOs, informing them of a new available donation. NGOs can view the donation details, assess suitability based on logistics, capacity, and need, and then decide whether to accept or decline the offer. This mechanism ensures efficient and transparent communication between donors and NGOs without manual intervention.

4. NGO Accepts Donation

Once an NGO decides to accept the donation, the system records this action and updates the donation status to "Accepted" in the database. This confirmation triggers a synchronization across both users' dashboards, enabling them to proceed with coordination and planning. At this stage, both the Organizer and NGO have access to the same donation details, including contact information and location data necessary for logistical planning.

5. Live Chat / Coordination (Built-in Messenger)

Upon acceptance, the system activates the built-in chat and coordination module, allowing real-time communication between the Organizer and NGO. Through this feature, they can: Confirm pickup timing, Discuss transport arrangements, Resolve queries or special requirements. This messaging system reduces miscommunication and enhances coordination efficiency, ensuring that food collection is completed smoothly and on schedule.

6. Food Collection

Once the communication is complete and arrangements are finalized, the NGO or its volunteers proceed to collect the food from the organizer's specified location. The collection details may be logged in the system, updating the status from "Pending" to "Collected." This step bridges the digital platform with real-world action, ensuring traceability and accountability.

7. Status Update: Donation Completed

After successful food collection, the NGO updates the donation status to “Completed.” This final update closes the loop of the process, marking the transaction as fulfilled in the system’s database. Both the Organizer and NGO can view this completion record on their dashboards, and automated acknowledgments or digital receipts may be generated for the donor’s reference. This transparency not only enhances user trust but also allows NGOs to maintain organized records for reporting and analytics purposes.

Chapter 6

Experimental Setup

6.1 Hardware Requirements

The experimental platform comprises a class of mid-range personal computing devices selected to represent typical end-user and development machines encountered in small and medium-scale deployments. Processors must be at least fourth-generation Intel Core i5 or i7, or an equivalent AMD Ryzen 5/7 series unit, providing a baseline of four physical cores, eight logical threads, and clock speeds between 2.4 GHz and 3.8 GHz. Random access memory of no less than 8 GB ensures adequate capacity for concurrent processes such as integrated development environments, emulators, and background services. Storage must consist of either a 500 GB hard disk drive or a 256 GB solid-state drive; while HDDs incur greater access latency, they offer greater capacity for large media assets, and SSDs deliver faster boot and application loading times. The operating system layer supports Windows 10, macOS Catalina (10.15) or later, and mainstream Linux distributions (e.g., Ubuntu 20.04 LTS), guaranteeing compatibility

with development toolchains and device-level dependencies such as Android emulators or native package managers.

6.2 Software Requirements

To realize the proposed architecture, the software stack integrates cross-platform frameworks, cloud services, and collaborative version control to accelerate development and ensure robust backend services. All software components adhere to current stable release cycles to leverage recent security patches and performance enhancements.

Integrated Development Environment: Android Studio (version 4.2 or later) and Visual Studio Code (version 1.50 or later) serve as the primary code editors and IDEs. Android Studio provides the Android SDK Manager and built-in emulators for native testing, while Visual Studio Code, augmented by the Dart and Flutter extensions, enables rapid prototyping and fine-grained code navigation across both front-end and back-end modules.

Frontend Framework: The user interface is implemented using Flutter, Google's open-source UI software development kit written in Dart. Flutter's reactive framework and widget library facilitate the creation of natively compiled applications for mobile, web, and desktop from a single codebase. This architecture promotes high performance through Ahead-Of-Time (AOT) compilation, and allows sophisticated animation sequences integral to the application's animated dashboard and micro-interaction components.

Backend Services: Firebase underpins the application's backend, offering a suite of serverless modules that obviate the need for dedicated server maintenance. Specifically, Firebase Authentication manages secure user login flows and integrates seamlessly with Flutter's authentication APIs. Firebase Cloud Functions provide on-demand execution of server-side logic, enabling tasks such as push notification triggers or dynamic email templating to occur in response to Firestore database events.

Database: Cloud Firestore, Firebase's NoSQL document database, persists application data including user profiles, donation listings, and message threads. Firestore's real-time

synchronization capabilities ensure that updates propagate automatically to connected clients, fulfilling the application's requirement for real-time chat and live donation feeds. Data modelling in Firestore uses collections and documents; for example, a "donations" collection contains documents keyed by unique identifiers, each with nested fields for item name, quantity, location, and timestamp.

Version Control: All source code, configuration files, and documentation reside within a Git repository hosted on GitHub. Branching strategies follow the GitFlow workflow, with feature branches branching from 'develop' and pull requests subject to peer code review prior to merging. GitHub Actions pipelines automate continuous integration tasks including static analysis (using dart analyze), widget testing, and build artifact generation for Android and iOS. Semantic versioning tags facilitate release management and deployment traceability.

6.3 Virtualization and Emulation

Testing encompasses both physical devices and emulated hardware. Android Virtual Devices (AVDs) replicate various screen sizes and Android API levels (minimum API 21, targeting API 30). Flutter's flutter driver integration tests operate on emulators to validate UI workflows, whereas unit tests execute in a headless environment. On macOS hosts, iOS Simulator (Xcode 12.5 or later) is employed to assess platform-specific behaviors. Continuous integration jobs spin up headless Linux containers to run Dart unit tests and Flutter driver scripts, ensuring consistency across development and CI systems.

6.4 Networking and Security Configuration

During development, the Firebase project is configured with CORS rules and security rules specifying that authenticated users may read public donation listings but only update their own data. HTTPS is enforced for all REST API endpoints, and Firestore rules require token validation via Firebase Authentication. API keys and service account credentials are stored in secure environment variables and accessed via Flutter's flutter_dotenv plugin, thereby preventing accidental exposure in public repositories.

6.5 Performance Profiling Tools

Performance optimization leverages Flutter's DevTools suite, including the Dart profiler, Timeline, and Widget Inspector to identify rendering bottlenecks and memory leaks. Network traffic is monitored with Charles Proxy, enabling inspection of HTTPS payloads between the Flutter client and Firebase endpoints. Android Studio's CPU and Memory profilers complement this by tracing native library calls and GC events on Android emulators. These tools collectively support iterative performance tuning of UI rendering, database queries, and function cold-start latencies.

Chapter 7
Results and Discussion

This chapter presents the outcomes of the NGO Connect Application and analyses its effectiveness across key functional stages, including event food listing, NGO request handling, real-time communication, and overall coordination between stakeholders. The results validate the system's ability to streamline food redistribution processes and enhance collaboration among event organizers and non-governmental organizations.

7.1 Results

The project outcomes confirm the successful implementation and functionality of the platform's core modules event management, food listing, NGO registration, and food request coordination. The developed system efficiently bridges the gap between food donors (and NGOs by offering a unified digital platform for communication and transaction transparency.

The integration of Firebase for authentication, database management, and push notifications significantly improved system reliability and real-time performance. The Google Maps API facilitated accurate location tracking and optimized NGO - event matching based on proximity, ensuring faster response times and reduced food wastage.

Testing across various scenarios demonstrated that the app provides a smooth user experience, with minimal latency during operations such as posting food availability or processing NGO requests. The use of a centralized cloud-based architecture enhanced data consistency and scalability, allowing multiple users to interact simultaneously without performance degradation.

The feedback from test users, including sample event organizers and NGOs, highlighted the system's potential for practical deployment. Most users appreciated the ease of navigation, transparency in food allocation, and the direct communication feature, which fostered trust and accountability in the redistribution process.

Overall, the results confirm that the NGO Connect App successfully meets its primary objectives: reducing food waste, promoting social welfare, and creating a collaborative ecosystem that connects surplus food donors with organizations serving the needy. The project demonstrates that technology can effectively be leveraged to address humanitarian challenges by ensuring efficient, transparent, and scalable food redistribution.

7.2 Demonstration

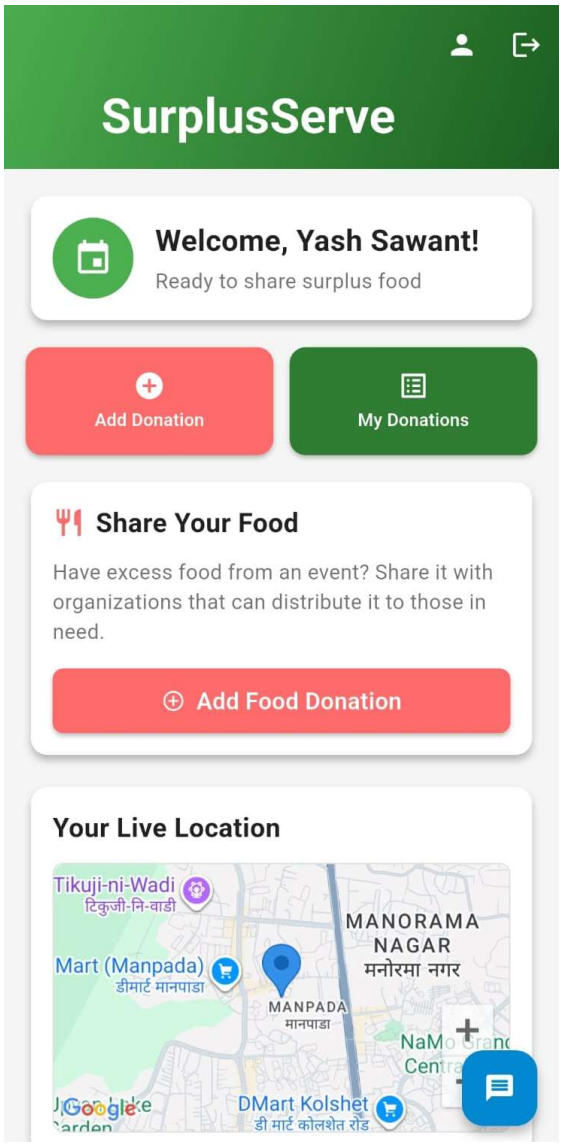

Fig 7.1 Event manager interface

The SurplusServe home screen welcomes the user, showing options to Add or view My Food Donations and displaying the location of the user.

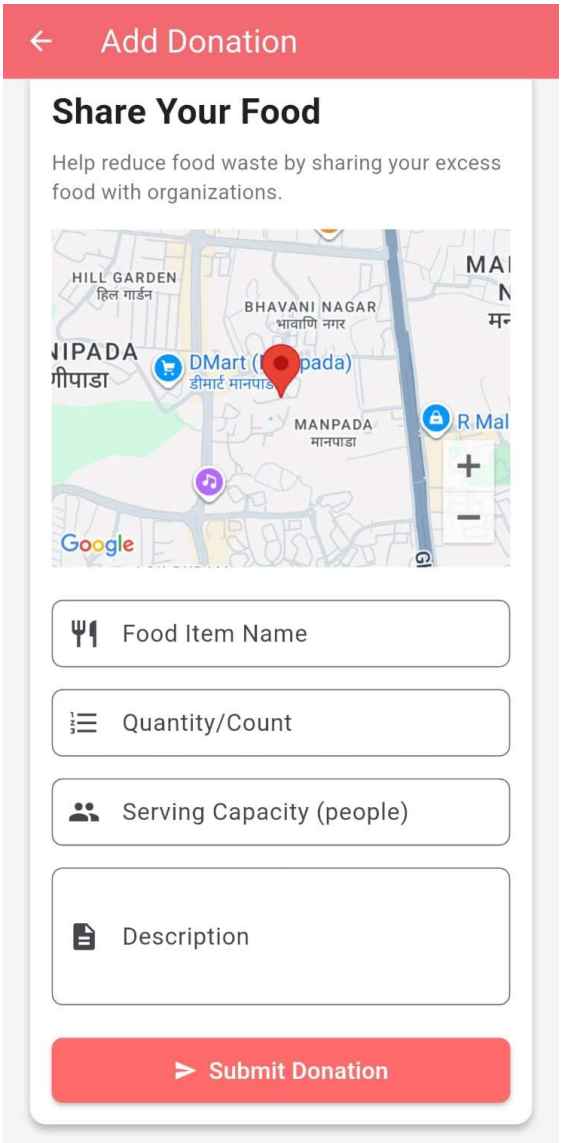

Fig 7.2 Add Donations Interface

This "Add Donation" screen is for sharing food, requiring the user to input the item name, quantity, serving capacity, and a description.

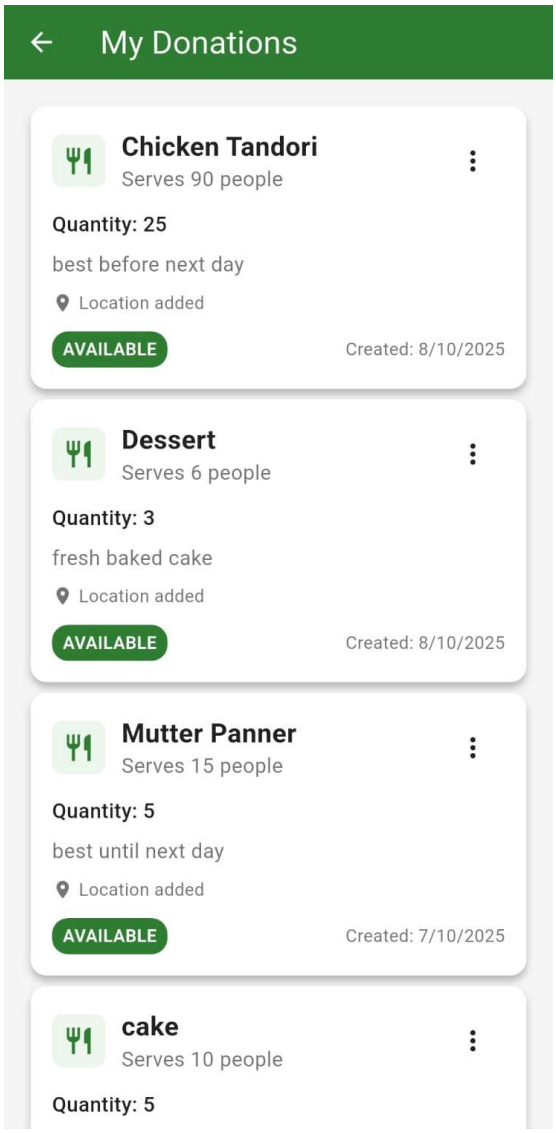

Fig 7.3 My Donations Interface

This mobile screen is the "My Donations" list, showing food items with their quantity, availability status and creation date

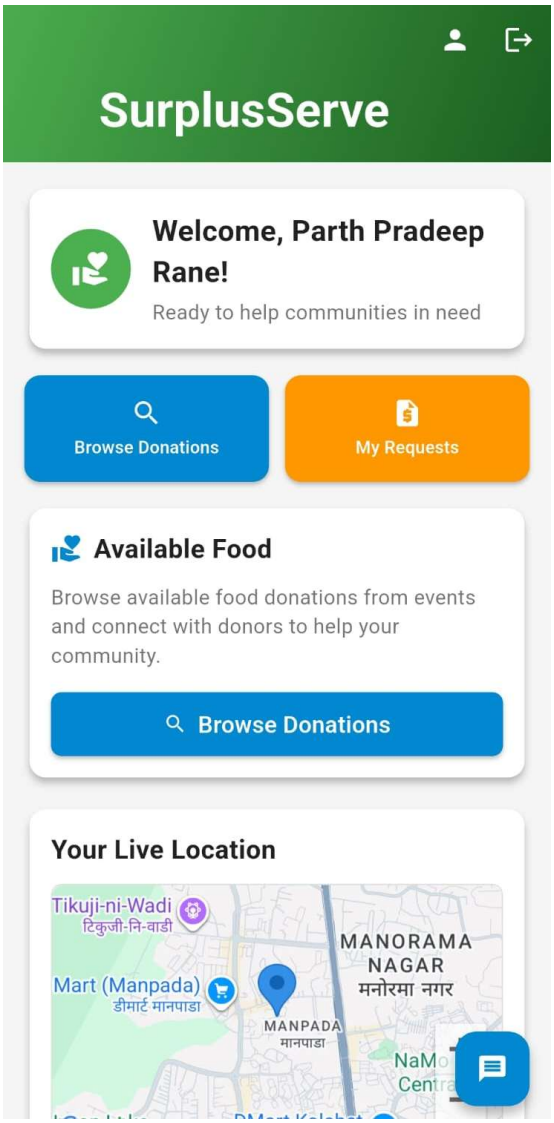

Fig 7.4 NGO User Interface

This is the SurplusServe home screen for user, showing options to Browse Available Food Donations, view My Requests, and see the user's Live Location.

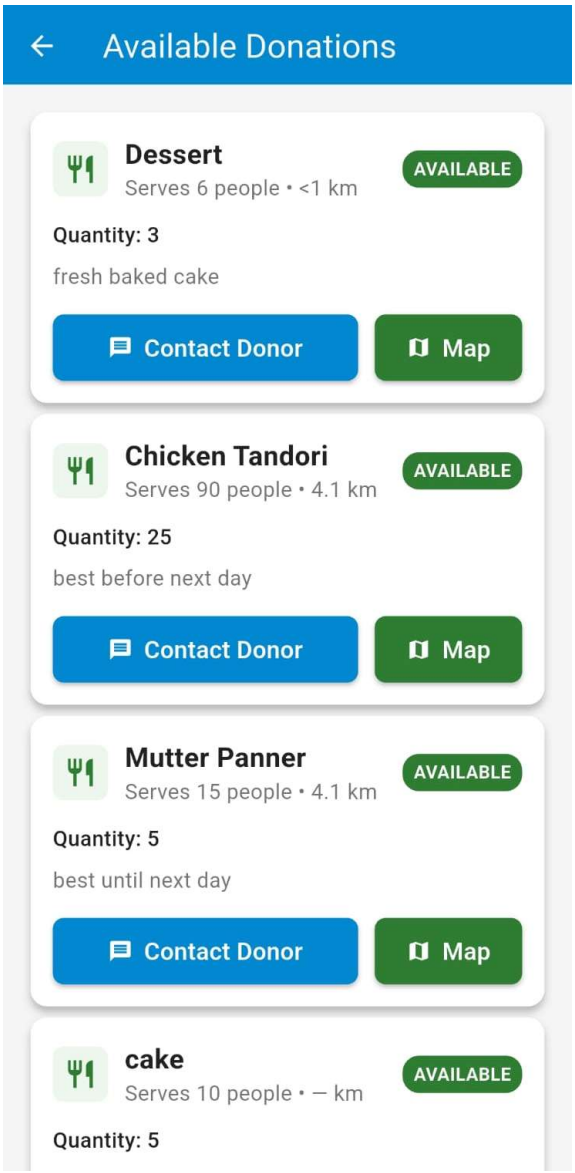

Fig 7.5 Available Donations Interface

This "Available Donations" screen lists food items, shows quantity, distance, availability and options to Contact Donor or view a Map.

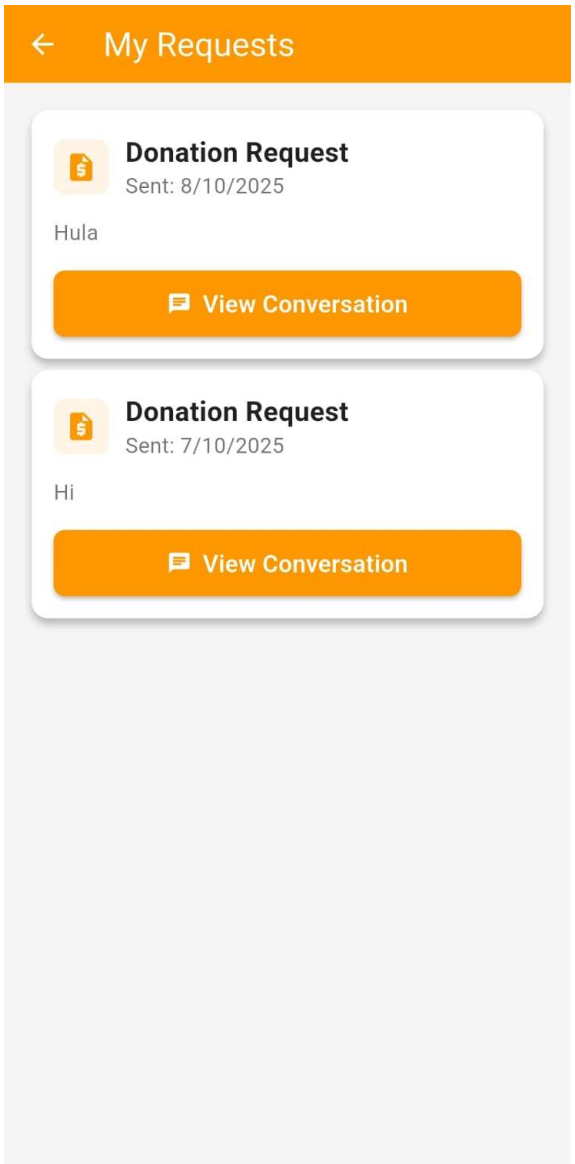

Fig 7.6 Request Interface

This "My Requests" screen lists separate Donation Requests sent along with the timestamps, each with an option to View Conversation.

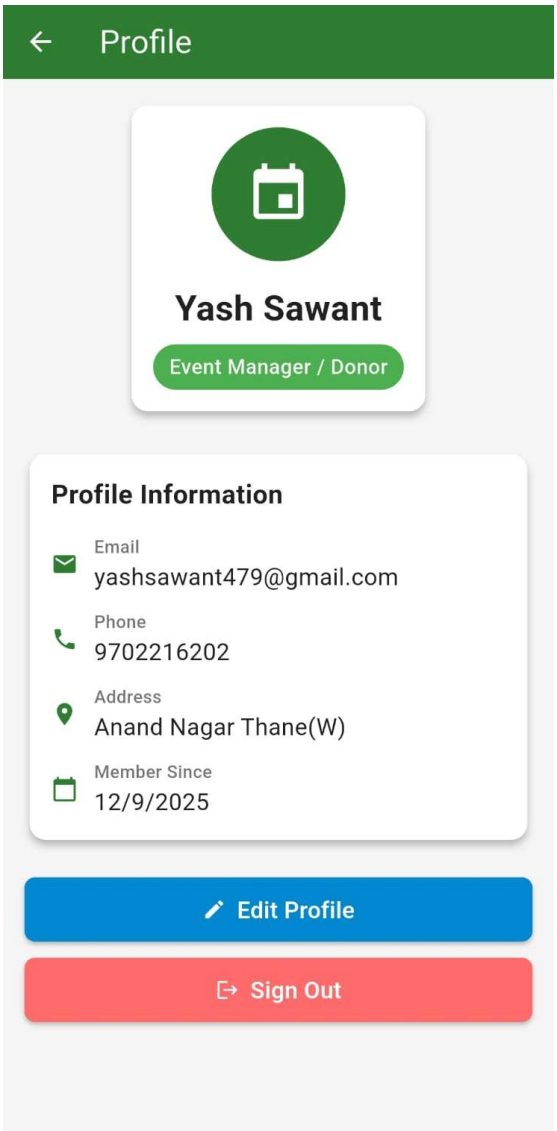

Fig 7.7 User Profile

This is a the app profile screen for user, showing his email, phone, address, member status, and options to edit or sign out.

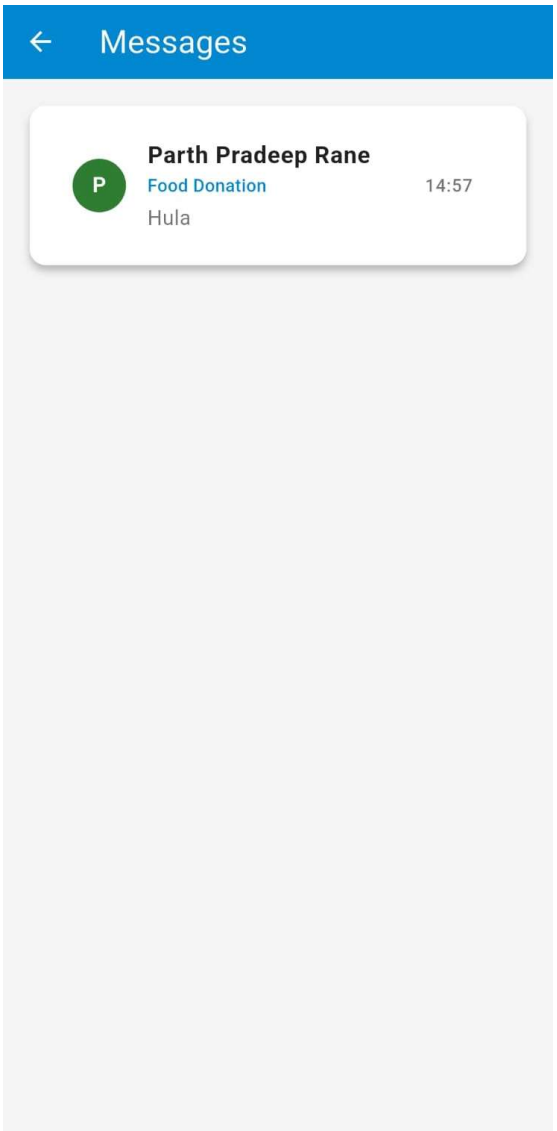

Fig 7.8 Messaging Interface

This Messages screen shows a conversation from other users regarding a Food Donation, with a preview of the last message with the timestamp.

7.3 Discussion

The results of the **"SurplusServe – Bridging Feasts"** validate the effectiveness of integrating Firebase, Google Maps API, and real-time communication mechanisms to facilitate efficient coordination between event organizers and NGOs for food redistribution. The discussion focuses on the system's key performance achievements and how its design choices address the limitations of traditional donation management and coordination methods.

Key Performance Outcomes

- **Streamlined Food Donation Process :** The core challenge of connecting event organizers with NGOs in real time was effectively resolved by integrating Firebase for seamless data synchronization and cloud storage. This integration enabled quick listing of surplus food items, instant request handling, and secure user authentication ensuring both data integrity and rapid updates across devices.
- **Location-Based Matching and Coordination :** Using the **Google Maps API**, the system automatically locates nearby NGOs based on event coordinates, allowing organizers to reach the most suitable recipients. This feature minimizes transportation time and reduces the risk of food spoilage, marking a major improvement over manual search or phone-based coordination.
- **Real-Time Notifications and Communication :** The inclusion of **Firebase Cloud Messaging (FCM)** ensures instant notifications whenever new food listings or NGO requests are made. Additionally, the in-app chat system allows donors and NGOs to coordinate logistics efficiently, eliminating delays caused by traditional offline communication.
- **Transparent and Traceable Transactions :** Each donation is logged and visible to both parties, providing a transparent chain of custody from donor to NGO. This promotes accountability and trust, solving a major limitation of existing manual systems where food donations are often undocumented or unverified.
- **Scalable and User-Friendly Architecture :** The use of **Firebase's Realtime Database** and cloud-hosted backend ensures that the platform remains scalable as the

number of users grows. The simple yet intuitive **Flutter-based interface** provides easy access for users with minimal technical expertise, ensuring inclusivity and wide adoption potential.

Overcoming Existing Limitations

The proposed system directly addresses several limitations identified in conventional food redistribution practices and existing applications:

- **Fragmented Communication Channels :** Unlike existing systems where coordination happens through disconnected platforms, NGO Connect centralizes all activities from listing to delivery confirmation within one unified app.
- **Lack of Real-Time Visibility :** The real-time database and notification features overcome the issue of delayed updates, ensuring that NGOs are immediately informed of available food, improving response speed, and reducing waste.
- **Inefficient Resource Allocation :** Through its location-based sorting and NGO categorization, the app ensures equitable food distribution based on proximity and NGO capacity, leading to better resource utilization.
- **Limited Transparency in Manual Systems :** The system's record-based approach ensures every transaction is logged and traceable, addressing the transparency and accountability issues commonly faced in manual donation models.
- **Difficulty in Scaling Operations :** Since the backend is cloud-based, the platform can easily scale to include more users, cities, and NGOs without needing major infrastructure changes making it adaptable for future expansion in the future whenver needed.

Chapter 8
Conclusion and Future Work

8.1 Conclusion

The NGO Connect App marks a significant step toward digital transformation in the nonprofit sector by uniting donors, volunteers, and NGOs on a single, cloud-based platform. It effectively addresses challenges such as fragmented communication, inefficient coordination, and lack of transparency in traditional NGO operations. Through a modular, role-based design, the app creates an integrated ecosystem where each stakeholder donor, volunteer, and NGO administrator interacts seamlessly to optimize the life cycle of donation and community service activities.

Developed using Flutter, the app achieves high performance and consistent cross-platform functionality, enabling dynamic dashboards, real-time tracking, and responsive user

experiences across Android and iOS. Complementing this is Firebase, which provides backend services for real-time data synchronization, secure authentication, and automated notifications. These technologies ensure reliable performance, data integrity, and minimal maintenance overhead.

The system follows the Model–View–Controller (MVC) architecture, ensuring a clear separation between interface, logic, and data layers, which simplifies testing, scalability, and maintainability. This structured approach enhances development efficiency and safeguards sensitive data such as donation records and user credentials.

In practical terms, the NGO Connect App can significantly improve operational efficiency in charitable organizations by reducing administrative workload, expediting coordination, and providing transparent impact analytics. Future developments could integrate machine learning for predictive volunteer matching and blockchain for transparent fund tracking, further demonstrating how modern technologies can drive social impact and operational excellence in the nonprofit domain.

8.2 Future Enhancements

Future upgrades of the NGO Connect App will focus on expanding functionality, scalability, and accessibility. An AI-driven donor prediction system will be introduced to analyze behavioural data and forecast potential donors, allowing NGOs to plan targeted fundraising campaigns efficiently. Similarly, an NLP-based chatbot will automate volunteer coordination, manage schedules, and respond to routine queries, reducing manual communication burdens.

To promote large-scale collaboration, the system will integrate a permissioned blockchain layer (using frameworks like Hyperledger Fabric) for transparent tracking of donations and resource distribution among partner NGOs. This decentralized approach ensures data immutability, accountability, and real-time auditing.

Additionally, multilingual support will be implemented using localization and translation frameworks to make the app accessible to diverse linguistic communities, encouraging broader

participation. Collectively, these enhancements will transform the NGO Connect App into a scalable, intelligent, and inclusive platform that empowers global humanitarian collaboration.

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